

Security and Privacy in Online Social Network

Lab 1 – Investigating the “AI Companion” Phenomenon

Due 6:00pm on 6/8/2026

0. POLICIES:

1. Coverage

Investigating the “AI Companion” Phenomenon: Data Collection and Analysis

2. Grade

This lab accounts for 40% of the final grade.

3. Individual or Group

This is an individual assignment; however, group discussion is allowed.

4. Academic Honesty

Violation of academic honesty may result in a penalty more severe than zero credit for an assignment.

5. Submission

Submit your code, collected data, and a report including data collection methodology, analysis, findings, and discussion, to course.zju.edu.cn.

6. Late Submission

Late submissions after 6:00pm on 6/8/2026 will NOT be graded.

1. Experiment Background and Objectives:

The AI companion phenomenon represents one of the most prominent socialization trends in artificial intelligence in recent years. It refers to situations in which users form emotional bonds, engage in deep conversations, and even develop “parasocial relationships” with virtual characters powered by large language models (LLMs). On Reddit, one of the largest overseas community forums, a substantial number of deeply engaged users have gathered (e.g., in subreddits such as r/MyBoyfriendIsAI, r/CharacterAI, r/replika) and shared extensive usage experiences, emotional appeals, and technical explorations.

This lab aims to explore users’ deep-seated perceptions of and interaction patterns with LLMs as “AI companions” through the collection and analysis of real Reddit data.

Upon completing this lab, you will master fundamental research methods in social media data collection, data cleaning and filtering, and text and thematic analysis.

2. Experimental Procedure:

Step 1: Determine your Research Question

Scientific research begins with a clear question. Select **one or more** questions from the table below as your research questions and develop a data retrieval strategy accordingly.

	Research Question	Problem Description
1	In what vulnerable situations do users seek support from AI companions?	Analyze the real-world factors (such as social anxiety or unemployment) that drive users to seek support from AI companions.
2	How do users evaluate the AI's short-term and long-term memory capabilities?	Analyze users' positive (coherence) and negative (privacy concerns functional failures) feedback regarding the AI companion's memory capabilities.
3	How does the security review mechanism influence interactions between users and AI companions?	Study how users perceive the impact of platform compliance audits on their sense of "closeness."
4	How do users attempt to breach the security mechanisms of AI companions?	Examine the prompt engineering techniques shared by users to see how they guide their AI companions to bypass security filters.
5	What negative emotions do AI companion updates trigger among users?	After analyzing the AI companion update, users experienced a profound sense of loss—even exhibiting behaviors resembling real-life breakups.
6	Is the user concerned that conversation data may be used for training?	Analyze the level of users' concern about data security—whether they are "extremely vigilant" or "completely indifferent."

7	What impact does the AI companion have on users' real lives?	The analysis examines whether AI companions enhance social interaction or contribute to social withdrawal in real-life settings.
8	How do users choose AI companion products?	Analyze the factors driving users when selecting AI companion products. Which factors lead users to switch between different AI companions?

Your task: (1)-(3)

- (1) **Select research questions:** Choose one or more from the above 8 questions.
- (2) **Identify the target subreddit:** Based on your question, find one or more of the most relevant Reddit communities. For example: r/MyBoyfriendIsAI, r/CharacterAI, r/replika, etc.
- (3) **Define keywords:** To achieve precise retrieval, convert your research question into search keywords. For example, for “Question 5 (AI Companion Update),” potential keywords include: update, changed, miss the old, goodbye, keep, etc.

Step 2: Data Collection and Filtering

This step requires you to obtain raw data from Reddit and clean it to produce a valid dataset that meets the requirements. You must retain at least 100 posts or long comments that are highly relevant to your research question.

Your task: (1)-(3)

(1) Select a data collection tool

Based on your technical background, it is recommended to choose one of the following methods to obtain data (please explain your choice in the report):

- **Method A: PRAW (Python Reddit API Wrapper).** You need to register an account on Reddit to obtain an API Key, after which you can perform searches using Subreddit and keywords.
Reference link: <https://praw.readthedocs.io/en/stable/index.html>
- **Method B: Third-party open-source databases/datasets.** Use an archiving tool similar to Arcticshift to retrieve posts within a specified time range.

Reference link: <https://arctic-shift.photon-reddit.com/>

- **Method C: Web crawler.** Utilizes Python libraries such as BeautifulSoup, Selenium, or playwright to directly scrape Reddit web pages.

(2) Save the collected data

The data you crawl must be stored in a structured format (CSV or JSON is recommended). Your dataset should include at least the following fields:

- ✓ Post_ID (Unique Identifier)
- ✓ Title (Post Title)
- ✓ Body/Text (Post or comment body)
- ✓ Timestamp (Release Time)
- ✓ User (Poster)
- ✓ Subreddit (Source Section)

(3) Data Screening and Cleaning

The raw data contains a significant amount of noise. You need to record and apply your “Inclusion/Exclusion Criteria”:

- ✓ **Initial filtering:** Remove posts with too few words (e.g., meaningless complaints consisting of just one or two sentences), advertisements, or pure image posts.
- ✓ **Manual verification:** Review the preliminary screened data to ensure that at least 100 remaining records provide sufficient information to support your research question in Step 1.

Step 3: Data Analysis and Research Analysis (Data Analysis)

After obtaining over 100 high-quality texts, you need to analyze them. Don't just list the data; instead, “refine” the key insights.

Your task:

You need to combine the following methods to analyze your data and answer your research questions:

- ✓ **Thematic Analysis:** Review all 100 data points thoroughly and assign tags to user statements. For instance, when analyzing the “security review mechanism,”

tags might include “feeling betrayed,” “thinking AI has become less effective,” or “seeking alternatives.” Identify the most frequent tags and distill them into 3–5 core themes.

- ✓ **Reference Support:** Select representative user quotes to substantiate your analytical conclusions.

You can also use other methods that you consider appropriate.

3. Final submission content and format requirements:

Before the deadline, package the following three items into a single ZIP file and submit it with the naming format: StudentID_Name_OSNLab1.zip.

(1) Original and cleaned datasets (Data): Submit a.csv or.json file. The file must contain at least 100 valid entries selected through final screening, including all data fields required in Step 2.2.

(2) Source Code (Code): Submit the script you use for data crawling, data cleaning, or analysis (.py or.ipynb Jupyter Notebook file).

(3) Experimental Research Report (Lab Report): Format: PDF document.

The report structure must include the following sections:

- **Research Question and Background:** Which question did you choose? Why did you select this question?
- **Data Collection Method:** What tools did you use (e.g., PRAW/crawlers)? What were your target Subreddit and search keywords?
- **Data Screening Process:** How much raw data did you collect? How did you filter out the final 100+ valid data entries? (Please describe your screening criteria.)
- **Results and Analysis (Core Section) :** Based on your data analysis, address your research questions. Use charts (e.g., posting time trends, keyword frequency, pie charts, etc.) to support your findings, and cite 3–5 representative user texts as evidence.

For report writing, please also refer to the *report template* and the *example report*, provided along with this assignment.